Multivariate Linear Regression Using California Housing Dataset

Dataset: California Housing Dataset – 20,640 records and 8 numerical input features.

Sample Data (5 rows)

MedInc	HouseAge	AveRooms	AveBedrms	MedHouseVal
8.3252	41.0	6.984126984126984	1.0238095238095235	4.526
8.3014	21.0	6.238137082601054	0.9718804920913884	3.585
7.2574	52.0	8.288135593220339	1.073446327683616	3.521
5.6431	52.0	5.817351598173516	1.0730593607305936	3.413
3.8462	52.0	6.281853281853282	1.0810810810810811	3.422

Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error

# Load the complete real dataset
data = pd.read_csv("california_housing.csv")

# Select multiple input features and target
features = ["MedInc", "HouseAge", "AveRooms", "AveBedrms",
            "Population", "AveOccup", "Latitude", "Longitude"]

X = data[features]
y = data["MedHouseVal"]

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train the multivariate linear regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict test values
y_pred = model.predict(X_test)

# Evaluate the model
r2 = r2_score(y_test, y_pred)
```

```
rmse = mean_squared_error(y_test, y_pred) ** 0.5

print("R2 Score:", r2)
print("RMSE:", rmse)

# Predict house value for new data
new_house = [[5.0, 20, 6.0, 1.0, 1000, 3.0, 34.0, -118.0]]
prediction = model.predict(new_house)

print("Predicted House Value:", prediction[0])
```

Sample Output

```
R2 Score: 0.5758
RMSE: 0.7456
Predicted House Value: 2.3498 (in $100,000 units)
```

Graph

